Adept: technical tools declaration

Updated 30 October 2026 against the supplied finalist **Technical & Tools Declaration** template. Its stated deadline is **1 November 2026**. Sections 1-8 follow that template. This document records repository evidence and explicit unknowns, not a signed compliance claim. Personal contact details belong in the private submission PDF, not in this public repository.

Updated 30 October 2026 for the root-only, single-public-server source revision, current dependency locks, live-data connectors and separate AI services. A source pin is not a claim about the inventory of an older deployed image.

1. Team and project overview

Template field	Response
Team name	Pecel Lele (repository team record; confirm registration spelling).
Project / solution name	Adept.
Name (leader)	Team to complete privately in the submission PDF.
Email and phone (leader)	Team to complete privately in the submission PDF.
University affiliation(s)	Xiamen University Malaysia, per the project README.
Live prototype URL	adept-my.replit.app: public pages and API/database health returned 200 on 24 October 2026. The deployed commit and the success of a newer publish attempt remain unverified. Local development uses :3000 and private API docs :8000/docs.
Code repository URL	pecel-lelele/adept; public visibility verified; confirm access again before submission.

2. Architecture and technology stack

2.1 System architecture overview

Adept keeps product source under root src/, tests under tests/, and documentation and submission material at their required root paths. One public Next.js standalone server renders pages, runs server actions and forwards API requests to private FastAPI on loopback. The root Python launcher checks configuration and API/database readiness, supervises both children, and performs no startup migrations or seeding. HTTP-only signed cookies hold local account sessions; PostgreSQL stores profiles, memberships, plans, evidence and catalogue data.

The batch research pipeline calls a pinned 384-dimensional MiniLM encoder on RunPod and validates dated trend snapshots before atomic publication. Optional 768-dimensional Gemini embeddings map unfamiliar skill phrases to catalogue suggestions. Matching and pathways use deterministic rules. Adi can use GLM-5.3 with Gemini failover to select validated answer-block IDs; displayed wording comes from server-authored facts.

Public employer/remote-job feeds, sourced courses, certification fees, official UN SDGs and BNM exchange rates complement research signals. GitHub Actions runs quality checks, database recovery tests, the actual root deployment smoke, weekly source audits and conditional trend refresh. Local Compose provides PostgreSQL only. The public site's reachability and MiniLM inference were observed separately; latest-source deployment and successful new snapshot publication require their own verification.

Diagram appendix: current system block diagram, full Mermaid collection, and the offline pan-and-zoom viewer. Diagram relationships are traced to routers, the root launcher, Next.js routes, FastAPI routers and migrations. Schema excerpts distinguish enforced foreign keys from logical joins.

2.2 Languages, frameworks and libraries

Versions below describe the reviewed source/lockfiles, not every deployed environment. The authoritative inputs are web

package, pnpm lock, Python project, uv lock, and worker requirements. Runtime fallback requirements remain in requirements.txt. Full transitive notices must be retained with distribution.

Name & Version	Purpose in Project	License
Python >=3.11 (deployment runtime 3.11)	API, pipelines and root supervisor	Python Software Foundation License
Node.js 24; pnpm 10.29.3	Next.js runtime and package management	MIT
Next.js 16.4.2; React and React DOM 19.2.9	Server rendering, routing, UI and same-origin API proxy	MIT
TypeScript ~5.9.3	Strict typing	Apache-2.0
Tailwind CSS / @tailwindcss/postcss 4.1.18	Styling	MIT
next-intl 4.14.9	English, Bahasa Melayu and Chinese	MIT
Three.js 0.180.0	Adi's interactive 3D mascot	MIT
lucide-react ^1.61.0	Interface icons	ISC
class-variance-authority 0.7.1	Component variants	Apache-2.0
clsx 2.1.1; tailwind-merge 3.3.1	CSS class composition	MIT
Google Fonts: Inter via @fontsource-variable/inter ^5.3.0	Installed Inter font assets; current global UI styling uses system Arial, not a live Google Fonts request	MIT package; SIL OFL 1.1 font
openapi-fetch 0.14.1; openapi-typescript 7.9.1	Typed API client and generated contracts	MIT
@types/node 22.20.5; @types/react 19.2.18; @types/react-dom 19.2.7; @types/three 0.180.0	Development type definitions	MIT
FastAPI >=0.161.1	HTTP API and OpenAPI	MIT
Pydantic >=2.7,<3; pydantic-settings >=2.16.1	Contracts and configuration	MIT
Uvicorn >=0.54.0	Private ASGI service	BSD-3-Clause
psycopg[binary] >=3.3.6; psycopg-pool >=3.3.3	PostgreSQL access and pooling	LGPL-3.0-only; bundled binary components have additional notices
python-multipart >=0.0.32	Multipart CV uploads	Apache-2.0
Python standard library	Signed local sessions and scrypt password hashing	PSF-2.0
PyMuPDF >=1.24.0,<2	In-memory PDF text extraction; review distribution obligations before release	AGPL-3.0 or Artifex commercial licence
HTTPX >=0.28.1	External API and model-provider requests; worker pins 0.28.1	BSD-3-Clause
PostgreSQL 16; pgvector (pgvector/pgvector:pg16; extension version not pinned)	Relational storage; separate 384-dimensional MiniLM and 768-dimensional Gemini indexes	PostgreSQL License
sentence-transformers 6.2.0; Transformers 5.20.2; RunPod Python SDK 1.12.0	Pinned MiniLM Serverless worker source; deployed image versions require separate verification	Apache-2.0 for Sentence Transformers/Transformers; MIT for RunPod SDK
PyTorch 2.14.1	MiniLM worker source dependency	BSD-3-Clause
pytest >=9.1.1; Ruff >=0.17.2; ESLint 9.39.5; eslint-config-next 16.4.2	Tests, quality checks and linting	MIT

uv	Python environment tooling; version is environment-dependent	MIT or Apache-2.0
Mermaid 11.12.0	Documentation-only diagram generation	MIT
PyYAML 6.0.3; actionlint 1.7.7	Workflow parsing and publication-policy validation	MIT
Prettier ^3.9.10	Development formatting	MIT
Playwright renderer	Documentation-only diagram rendering	Apache-2.0

The Express/Drizzle/generated-client/mockup artifact scaffold has been removed from this source revision. It is not part of Adept's product stack. Framework code remains Next.js; this is not a conversion to Vite or a static SPA.

Docker Desktop/Engine and Compose are local development tools under their applicable Docker distribution terms; they are not a cloud host. This draft does not assert that every transitive licence obligation has been discharged. Review PyMuPDF's licensing statement and upstream notices before signing the final compliance checklist. Adept's own repository licence is Apache-2.0; that does not replace dependency licences.

3. APIs and cloud services

3.1 External APIs

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
arXiv cs.AI RSS	Offline research-attention ingestion; historical replay windows also cite the query API. Current ingest uses RSS; no arXiv call on a recommendation request.	Public/free; rate limits and terms apply
UN Statistics SDG Goal and Target APIs	Goal and target sync; scheduled source audit	Public/free
Microsoft Learn Catalog API and exam-pricing page	Course/module metadata, catalogue sync, availability checks and certification prices	Public/free
Coursera courses API	Curated slug/workload availability checks; no subscription or sponsorship inferred	Public endpoint; provider terms apply
Bank Negara Malaysia OpenAPI	Exchange rates for MYR price estimates	Public/free
Public job-posting feeds: Greenhouse, Lever, Ashby, Workday, Jibe, Himalayas, Jobicy, Remote OK, Remotive	Dated vacancy and skill evidence; these nine source families occur in the committed 23 October extract. SmartRecruiters is excluded from configured sources.	Public access; each source's terms apply
Optional keyed job APIs: Careerjet, Jooble, Web3.career	Connectors exist but are skipped in the committed fetch log; run only when configured	Authenticated; provider terms and credentials required
Cited provider HTTPS pages (courses, certifications)	Course facts, link audits and certification-page reachability	Public page access; not a single API
RunPod: GLM-5.3 for Adi; Serverless MiniLM batch embeddings	Separate endpoints and credentials: Adi selects validated answer blocks; MiniLM generates offline 384-dimensional research/catalogue vectors	Server API keys; sponsor resource usage declared in section 5; provider terms and charges apply
Google Gemini API (3.5 Flash Lite, 3.6 Flash, 3.1 Flash Lite; Embedding 001)	Optional Adi failover in that order; optional 768-dimensional skill suggestions using redacted short phrases and an existing catalogue index	Server API key; optional provider integrations; configured failover requires credentials, not a blanket production-disabled claim

The BLS baseline is committed data, not an external API called by the running recommendation service. Pinned MiniLM weights are downloaded from Hugging Face while building the worker image; inference is hosted on RunPod. The worker loads weights offline at runtime. Do not include API keys in this document.

3.2 Cloud platforms and deployment

Service & Provider	Purpose in Project	Plan / Region
GitHub repository / Actions	Version control, quality CI, source audit, worker publication and conditional trend refresh. The first real refresh failed saving to private database host `helium`.	Public repository; GitHub-hosted Linux runners (`ubuntu-latest`)
Local Docker Compose	PostgreSQL/pgvector only; web/API run through root commands	Local Windows Docker Desktop; no cloud region
Replit Autoscale	Hosts the deployed public prototype at `https://adept-my.replit.app`; landing, About and API/database health returned 200 on 25 October. One public Next.js server, private FastAPI; health does not identify the Git revision.	Sponsor-provided Replit Core subscription; Autoscale publishing
Replit integrated PostgreSQL	Managed application database for profiles, catalogue, snapshots, plans, progress and organisations; development and production databases are separate	Replit's built-in managed PostgreSQL, attached to the Core project
RunPod Serverless / hosted inference	Deployed MiniLM worker for offline embeddings and separate GLM-5.3 endpoint for Adi. Successful inference does not establish successful new snapshot publication.	US$25 sponsor-provided credits; usage-based Serverless/inference
GitHub Container Registry (GHCR)	Stores/distributes the MiniLM worker container image	Public `ghcr.io/pecel-lelele/adept-minilm` image

The Core subscription, US$25 credit allocation and use of Replit's integrated PostgreSQL were confirmed by the team leader on 25 October 2026. Deployed authentication uses Adept's signed local accounts stored in the same Replit database. The Plan / Region column records confirmed plans/resources and does not infer physical hosting location from the team's location. For deployment planning, Replit **Asia** is the recommended geography for a Malaysian audience; RunPod's available-data-center scheduling is recommended for this offline batch encoder. These are recommendations, not verified active-region claims. See Replit geography, Replit database, and RunPod data-center settings.

Publishing needs the production database connection, stable session secret, shared proxy secret, verified forwarding-hop count and exact origin. They are supplied in hosting secret/config stores, never in the PDF. GitHub trend refresh separately needs `TREND_DATABASE_URL`, `TRENDS_RUNPOD_API_KEY` and `TRENDS_RUNPOD_ENDPOINT_ID`. Its latest observed historical run failed reaching the private development host `helium`; database preflight now rejects that configuration before inference. A skipped/green job is not evidence of snapshot publication.

4. AI tools, models and datasets

4.1 AI models used in the solution

Model and version	Source / provider	Licence / terms	Usage
sentence-transformers/all-MiniLM-L6-v2	Sentence Transformers weights hosted in the team's RunPod worker	Apache-2.0	Pre-trained, no team fine-tuning. Pinned revision `1110a243fdf4706b3f48f1d95db1a4f5529b4d41`; 384-dimensional sentence/skill embeddings, cosine threshold 0.55.
GLM-5.3	Runpod, model `glm-5.3`	Provider terms; team to review before signing	Grounded Adi assistance through an OpenAI-compatible endpoint; low reasoning and bounded output. Local request verified.
Gemini chat	Google; configured fallback list in section 3.1	Provider terms	Adi failover, separate credentials; tested fallback behavior is not a claim of a current live request.
Gemini embedding-001	Google Gemini REST API	Provider terms	Optional catalogue indexing/short skill-phrase suggestions, 768 dimensions. A separate purpose from MiniLM research embeddings; user confirmation required.

Six-stage trend pipeline: arXiv RSS and committed BLS-derived data → strip HTML and split abstracts → MiniLM sentence/

skill vectors and per-week mention frequencies → log-ratio research velocity blended 0.6/0.4 with the labour baseline, with scenario transformation → fresh-ingest, distinct-period, minimum-coverage, known-ID and finite-value validation → atomic publication of `trend_snapshots` and `trend_snapshot_meta`. A validation failure prevents new snapshot publication; ingestion audit/raw data may already have been written.

This is a semantic encoder plus heuristics, not a trained predictor of actual job vacancies. The fast-scenario range is not a calibrated statistical confidence interval. See scoring, publication, and exposure methodology.

4.2 Datasets

Counts below describe committed seed files inspected on 24 October 2026 (job rows last seen 23 October), not the contents of every deployed database. No model was fine-tuned on these fixtures.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Custom skill and role catalogue (140 skills, 31 roles, 28 targets)	Taxonomy seed	Team-curated / synthetic	Team curation; upstream references retained; not a complete O*NET/ESCO import	No
O*NET 31.0 occupation crosswalk	O*NET, curated provenance	Public-derived / curated	CC BY 4.0 with stated exceptions; credit and modifications required	No
arXiv cs.AI research signal	RSS ingest and cached windows	Public	arXiv terms; metadata and paper content have different rights; no blanket CC0 claim	No learner data; raw feed metadata may contain public author names
BLS 2023-2033 labour baseline	Committed extract, BLS	Public / team-derived	US government data, cite BLS; US baseline, not a Malaysian vacancy forecast	No
Canonical skill exposure and historical snapshots	Seed data and fixtures, methodology	Curated / public-derived	Reviewer-assigned exposure estimates, not probabilities; demo snapshots replay October 2024 research windows	No
17 SDG goals and 169 targets	UN SDG API, seed	Public	UN data terms and attribution	No
61 role-target and 165 skill-target links	Mapping seed	Team-curated	Draft team interpretations with cited evidence; no UN endorsement	No
Course records (98) and prerequisite graph	Catalogue sources, seed	Public facts / curated	Providers retain their terms; prices and HRD Corp notes are dated estimates	No
Job-posting extract (1,210 records, committed 23 October 2026)	Public job boards and employer endpoints	Public / curated	Each board's terms; public access is not an open-data licence; counts do not guarantee current availability	No learner data; postings may name public employers
Certification and exam-pricing extract (14 exams)	Microsoft, AWS, Google Cloud, IIBA and IAPP sources	Public / curated	Issuer site terms; dated prices	No
Exchange rates	BNM OpenAPI	Public	BNM terms and attribution	No
Synthetic personas, organisations, regions and CVs	Five selected demo CVs/profiles, regression fixtures, seeds	Synthetic	Team-created evaluation/test data. Selected CVs: Imran, Sofia, Meera, Adrian and Nur Hana; additional intake settings are editable demo assumptions, not extracted qualifications.	No real identities
OpenDOSM monthly Labour Force Survey	Official catalogue	Public national aggregates	CC BY 4.0; retain DOSM attribution; monthly observation dates differ from retrieval dates	No

MyCOL 2024/2025 occupation groups and reviewed role crosswalk	TalentCorp source and mapping evidence	Public-derived / curated	TalentCorp publication terms; no open-data licence inferred. 66 source groups, one reviewed stored role-ID mapping; historical badge only, no ranking effect	No

4.3 AI-assisted development tools

Yes. The team defined Adept's problem, product direction and SDG focus. Claude Code was the primary coding assistant and produced most of the application code, alongside its demo-video and pitch-design work. Other AI tools supported debugging, exploration and review. The following usage was confirmed by the team leader on 25 October 2026. Exact tool/model versions were not recorded; do not infer them from installed software or repository context files.

Tool & Version	Used For	Extent of Use (minor / moderate / major)
OpenAI Codex; exact app/model version not recorded	Repository inspection, debugging, resolving Greptile findings, code/configuration corrections, checks and documentation review	Major debugging and review support
Anthropic Claude Code; exact app/model version not recorded	Produced most application code; also supported demo-video creation and pitch-deck design concepts, layout and presentation material	Major; primary implementation assistant
Replit Agent; exact version not recorded	Exploring the repository and understanding its structure	Minor
Google Antigravity agent; exact version not recorded	Exploring the repository and understanding its structure	Minor
Greptile; exact version not recorded	Reviewing pull requests and identifying issues for the team to assess and resolve	Moderate

Open-source skills/plugins and connected-app integrations extended the team's authoring and inspection workflows. Declare the actual packages/repositories and relevant external services used, with their applicable terms; installing a tool does not establish that every capability or connected service was used. These extensions are development tools, not additional models embedded in Adept. The team retains responsibility for the submitted product and materials.

Review and responsibility (maximum 100 words):

The team chose Adept's problem, product direction and SDG focus. Claude Code produced most of the application code and supported the demo video and pitch design. Codex supported debugging and corrections; other tools assisted exploration and PR review. Team members assessed outputs and retained final decisions. Technical claims are checked against source, dependency records, datasets and test results. AI-generated material requires human review; passing tests does not prove universal fairness, legal compliance or employment outcomes. The leader reviews the final documents, confirms actual tool and sponsor usage, and takes responsibility for the submission.

4.4 Content and presentation tools

The team identified the following presentation tools on 30 October 2026. This inventory does not establish the contents of a final exported video. Retain the final asset credits and confirm which generated takes were used before signing this declaration. These tools are used for presentation production, not career recommendations in Adept.

Tool / version	Presentation purpose	Evidence and outstanding details
Google Fonts	Presentation typography where used	Confirm the exact exported font families and retain their applicable font licence notices.
Remotion 4.0.538	Motion graphics and demo-film composition	Version recorded by the team; the local video workspace is gitignored and not evidence of a reproducible application dependency.
Mixkit	Presentation media, music / sound effects as applicable	Record the exact downloaded asset pages and applicable item licences in final credits; these are not supplied in the tracked repository.
ElevenLabs	Demo/walkthrough video voiceover narration only (`submission/demo-video`)	The team confirmed use on 30 October 2026. Team recorded use strictly for demo video audio production with stock voice `21m00Tcm4TlvDq8ikWAM` (Rachel); no voice cloning; not integrated into or used by the Adept web application or recommendation runtime.

The demo's statistical context and dates are listed in the demo evidence register. WEF forecasts are global estimates; the Malaysian figure is a conditional ministerial warning, not observed job losses caused by AI. MiniLM is already declared in section 4.1; public job feeds, BNM rates and the O*NET crosswalk are declared in section 4.2. Workday career-site connectors use an undocumented public-facing interface; public reachability does not establish an open-data licence or a stable API contract.

5. Sponsor-provided resources

Yes: Replit deployment and RunPod inference/Serverless compute. The team confirms using Replit Core as the sponsor-provided subscription for deployment and the integrated PostgreSQL database. The team received US$25 in RunPod sponsor credits and uses RunPod for the deployed MiniLM Serverless worker and Adi's configured GLM-5.3 inference endpoint. These benefit details are confirmed by the team leader; the event site identifies Replit and RunPod as supporters. No subscription duration or mentoring participation is inferred.

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
RunPod	US$25 sponsor credits; GLM-5.3 inference endpoint and Serverless MiniLM worker	Adi's validated answer-block selection and offline public-research/catalogue embeddings. Worker inference succeeded; successful new trend-snapshot publication is a separate check.
Replit	Sponsor-provided Core subscription, cloud deployment and integrated PostgreSQL	Hosts the public Adept web/API and managed application database. Replit Agent was also used for repository exploration, as declared in section 4.3.

Essential or replaceable (maximum 100 words):

Replit hosts the deployed prototype and makes it accessible to judges. RunPod Serverless runs the MiniLM research encoder, while a separate RunPod endpoint supplies Adi's GLM-5.3 inference. These resources are important to the current deployment but replaceable: the web/API can move to another compatible host, MiniLM has a local CPU backend, and Adi has Gemini failover and a deterministic guide. Role matching and pathway generation use deterministic logic and saved data rather than requiring generative inference for each recommendation. Successful worker inference and successful publication of a new trend snapshot are separate checks.

6. Security, privacy and responsible use

6.1 Data handling and privacy

The demonstration uses synthetic identities, organisations and CVs. PostgreSQL can store account names/emails, scrypt password hashes, skills, preferences, memberships, learning plans, progress and evidence links. CV files are parsed in memory and not retained; confirmed skills are saved after review.

When enabled, Gemini receives short, contact-redacted skill phrases. Adi providers receive sanitised conversation/context and candidate answer blocks; do not claim that all personalised information stays local. MiniLM receives public research/catalogue text. PII filtering is a mitigation, not a guarantee.

Signed sessions use HTTP-only cookies. Organisation workspaces constrain sharing. The public proxy signs trusted visitor addresses; uploads/body sizes and rates are limited. Account export and password-confirmed deletion exist, including a last-manager safeguard. No automatic retention schedule or verified backup-erasure policy is implemented. Production requires HTTPS, managed secrets and operational review. No real personal data enters the competition demo.

6.2 Known limitations and risks

Research attention and US labour projections do not establish Malaysian vacancy forecasts. Job feeds supply dated openings, not guaranteed availability or employment outcomes. Curated catalogue coverage is limited; replay snapshots include historical observation windows. Scenario/exposure values are heuristic estimates, not calibrated probabilities. SDG links are team interpretations, not verified social impact.

CV parsing can miss aliases or scanned text; optional suggestions require confirmation. Provider-page checks do not establish certificate ownership. Fees, eligibility and upstream APIs change. Adi selects checked answer blocks, but relevance can still be wrong and human judgment remains necessary. Validation, source metadata, deterministic fallbacks, counterfactual checks, manual review and bounded network operations mitigate specific risks. Public reachability and green CI do not establish a deployed commit, security certification, universal fairness or improved employment outcomes.

7. Compliance and readiness

Final attestations remain unchecked until the team completes its review.

[ ] Every service, development tool, model and dataset used by every member is declared.

[ ] Dependency/data licences and required notices have been reviewed, including PyMuPDF.

[ ] No credentials are exposed; final repository and deployment inspection completed.

[ ] AI-assisted work has been disclosed and reviewed by the team.

[ ] Sponsor resources, including mentoring/credits outside source control, are confirmed.

[ ] Judges can access the public prototype and repository links in section 1.

[ ] Team fields and all final template answers are complete within their word limits.

8. Declaration

Unsigned draft. The leader must review actual team usage and complete the supplied accuracy declaration. A repository-generated draft cannot attest on the leader's behalf.

Field	Completion
Signature	Leader to sign privately after review.
Date	Actual signing date.
Final filename	`SDC2026_<RegisteredTeamName>_TechDeclaration.pdf`

Implementation references

Compose · requirements · web manifest · auth · LLM providers and failover · trend workflow · source audit · catalogue sources.

Final-round evidence update - 30 October 2026

The source declaration and this export distinguish runtime services from presentation tools. RunPod hosts the pinned all-MiniLM-L6-v2 encoder and the separate GLM-5.3 endpoint; Gemini is the separately authenticated Adi failover. Replit hosts the prototype and PostgreSQL. Google Fonts' Inter is distributed through the declared Fontsource package; current app styling uses Arial/system fallbacks. ElevenLabs was confirmed by the team as used, so it is declared as presentation audio tooling, not an Adept runtime model. See section 4.4 for the team's stock-voice record; the synthesis model is unrecorded.

Human planning-time KPI: **0 testers, 0 measured human trials**, as confirmed by the team on 30 October 2026. No achieved planning-time reduction is claimed. Automated local API measurements are reported separately in `docs/kpi-automated-journeys.md`; they exclude human review, browser interaction, network and a manual baseline. This remains an unsigned draft, with private team fields reserved for completion.